

DSC5101 In-Class Project

No collaboration is allowed. Submit by IVLE.

You are allowed to access IVLE and DSC5101 module's files there. You are not allowed to use any other website.

Please include your name and student ID on the submitted document(s). Further, include your student ID in the file name, for instance InClass_A1234567.doc, and zip the folder if you have multiple files.

If you need to make additional assumptions, write them down and explain them shortly.

1. Two separate questions:

a) Among uniform pricing, second degree price discrimination, and first degree price discrimination, which is the most profitable pricing policy and which is the least profitable pricing policy? Explain shortly. (2 points)

b) Two members of a criminal gang (A and B) are arrested and imprisoned. Each prisoner is in solitary confinement with no means of communicating with the other. The prosecutors lack sufficient evidence to convict the pair on the principal charge. They hope to get both sentenced to a year in prison on a lesser charge. Simultaneously, the prosecutors offer each prisoner a bargain. Each prisoner is given the opportunity either to: betray the other by testifying that the other committed the crime, or to cooperate with the other by remaining silent. The offer is:

(i) If A and B each betray the other, each of them serves 2 years in prison

(ii) If A betrays B but B remains silent, A will be set free and B will serve 3 years in prison (and vice versa)

(iii) If A and B both remain silent, both of them will only serve 1 year in prison (on the lesser charge)

Model this situation with a game model and solve the equilibrium. (3 points)

2. Use data Q2_data (in dta, csv or xlsx format) in the "Examples and projects" folder in IVLE and the corresponding "read me" file (Q2_readme.pdf) in the same folder.

State A amended its minimum wage law, increasing the minimum wage by 12%. The amendment went into effect on January 1, 1997, and it was the only revision during 1995 to 1998. Meanwhile, an adjacent state, State B, remained the minimum wage unchanged during the same period. The increase in the minimum wage might affect the employment in State A.

Evaluate the effect of the minimum wage increase on employment. Explain shortly your findings and your explanation for the results.

Include also your code (Stata, R, Matlab, SPSS, or other software). See above how to name the file (include your student ID in the file name). (5 points)